

Aayush Gupta

MIT EECS BS '22/MS '23, 4.8 GPA

RESEARCH

DINOHash: Adversarially Robust Perceptual Hashes ([CODEML: ICML '25](#))

- Started and led research effort into adversarially training alternative to Neuralhash
- Proposed better alternative to watermarks for image provenance via SFT on DinoV3

Interpretable Epigenetic Clocks to Infer Aging Mechanisms ([bioRxiv '20](#))

- First work to analyze multi tissue age prediction using gradient boosted trees
- Research in team of 3 under Elvira Kiniza in Manolis Kellis lab

Self-attention for Graph Neural Networks in Computational Chem (2019)

Won MIT Generator award for Best Project 2018

- Replaced message passing with query-key-value attention on QM9 dataset.

RL for Target Search with Drones (2017) ([arXiv](#), first author)

Published + presented at 2017 IEEE/RSJ Int'l Conf on Intelligent Robots and Systems (IROS)

- Improved rescue time by 3.3x with RL via a POMDP. Cited by 25
- Used Julia, SARSOP, and Python to build and test on a custom built drone. Led team of 3.

WORK EXPERIENCE

Proteus Cofounder ('26)

- Built multi-device computer use harnesses + models with tree based speedups and RL ([post](#))
- Build custom SDPO pipeline to do online RL on trajectories atop Qwen 3.6 + custom video encoder models built with direct video codec input

Alegria Neurotech ('25)

- Used ultrasound to invent writing fake smells to people's noses: [writetobrain.com/olfactory](#)
- Built 1.5 PB cluster for data factory, and built microbubble reconstruction from ultrasound

ZK Email Cofounder ('25)

- Invented new cryptography to verify redacted emails, scaled to 10 people and \$1M+ revenue
- Useful for whistleblowing, peer to peer exchanges, and redacted email verification

OpenPassport Cofounder ('24-'25)

- Used zero knowledge proofs to verify redacted passport data to protect user privacy
- Acquired by [self.xyz](#). Helped with technical architecture, funding, hiring, and acquisition.

Genesis Therapeutics: Molecular Machine Learning Team (2021)

- Researched new equivariant graph neural networks using cross products for drug prediction

NVIDIA AI Intern: Perception Team (2020)

- Used Tensorflow to build superresolution heads + correct model bias. Sped eval by 10x

Lipoker.io Cofounder (2020)

Created [lipoker.io](#), the first videochat poker site with no signup or downloads. Led team of 4-8.

- Created with Flask, SQLAlchemy, PostgreSQL, React, on GCP with Unicorn and Nginx.
- Grew to 10,000+ monthly sessions and partnerships with [gather.town](#) + others

Copysmith AI CTO/Cofounder (2020)

- Recruited + led team of 6 engineers and designers. Created MVP for 100+ users.
- Deployed generative NLP model in Pytorch with AWS and Cortex, scalable up to 100K users.

Scale AI Intern (2019)

- Modeled untrusted labelers for 95% confidence interval on bounding boxes
- Reduced LIDAR labeling errors by 8% via new incentive system in React, SQL, TS, Python

Auto-LaTeX Equations Creator (2015-)

Over **30,000,000** weekly users, 4.0+ star rating.

- Coded and branded a Google Docs add-on to render LaTeX equations

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AWARDS

USA Computing Olympiad National Camp Finalist (2016)

Top 28 pre-college competitive programmers vying for team USA spot. Mastered algorithms like dynamic programming, binary trees, and graph theory in C++.

Putnam Top 500 (2018)

USA Junior Math Olympiad (2016)

Top 200 of > 70,000 students.

USA Physics Olympiad Silver (2017)

Top 150 physics students in the USA.

North American Computational Linguistics Olympiad Finalist (2016)

Top 50 in USA.

Informal Coursework

Stanford's CS336

PROJECTS

Hack Lodge Organizer (2019)

MIT [Applied AI Labs](#) Founder ('20)

Healthcare AI in Sontag Lab (2021)

Securiti.ai Winter Intern (2019)

PLUME ECDSA Nullifiers ('22, 11 cites)

LANGUAGES

Python, Pytorch, JS, C++, Rust, Go, Julia

MIT CS COURSEWORK (4.8 GPA)

Graduate

6.824: Distributed Systems
6.871: ML for Healthcare
6.172: Performance Engineering
6.864: Graduate NLP
6.867: Graduate Machine Learning
6.438: Algorithms for Inference
6.890: Deep Learning + Algorithms
6.857: Cryptography and Security

Undergraduate

6.047: Computational Bio
6.036: Intro to Machine Learning
9.66: Computational Cogsci
6.033: System Design
6.046: Advanced Algorithms
6.041: Probability
18.100B: Real Analysis
8.041: Quantum Physics I
6.03: EECS for Medical Devices
6.004: Computation/Assembly

Founder/President of Saratoga High IoT Club (2016-18)

Attained \$5000 funding from Micron foundation.

Taught with Arduinos to engage students in laser tag, game bots, theremins.

Taught A-Star USA Computing Olympiad Camp (2017)

Managed classroom of ~20 A-Star USACO Silver students

Built in-home grading system for adversarial bots on final

[Other Info](#)